



KENYA INSTITUTE OF CURRICULUM DEVELOPMENT
A Skilled and Ethical Society

SENIOR SCHOOL CURRICULUM DESIGN

AGRICULTURE

GRADE 10

2025

First Published in 2025

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ISBN: 978-9914-52-911-1

Published and printed by Kenya Institute of Curriculum Development

FOREWORD

The Government of Kenya (GoK) is committed to ensuring that policy objectives for Education, Training and Research meet the aspirations of the Kenya Constitution 2010, the Kenya Vision 2030, National Curriculum Policy 2019, the United Nations Sustainable Development Goals (SDGs) and the Regional and Global conventions to which Kenya is a signatory.

Towards achieving the mission of Basic Education, the Ministry of Education (MoE) embarked on a reform process that culminated in the adoption of a Competency Based Education (CBE) system in 2017. This came after extensive research, stakeholder engagement international benchmarking and consideration of the 21st Century socio-economic realities. CBE in Kenya adopted a 2-6-3-3 structure where the numbers indicate the years learners take at various levels in Basic Education (Pre-Primary, Primary, Junior School and Senior School; respectively). CBE has for critical elements: The Competency Based Curriculum (CBC), Pedagogy (teaching and learning), Resources and the Competency Based Assessment. The MoE oversees the provision of resources while the other components (Curriculum, Pedagogy and Assessment are handled by the Kenya Institute of Curriculum Development, (KICD), Teachers Service Commission, (TSC) and the Kenya National Examinations Council (KNEC); respectively.

KICD has successfully developed curriculum designs that have been used in the implementation of the Competency Based Curriculum (CBC) for the Pre-Primary, Primary and Juniors School levels. The first cohort of learners pursuing the CBC will transit to Senior School (Grades 10-12) in 2026. The curriculum designs at this level are expected to facilitate learners to acquire relevant knowledge, skills, values and attitudes (competencies) in preparation for tertiary education and the world of work. I urge all agencies of the GoK and other stakeholders in Education to use the designs to plan for effective and efficient implementation of Competency Based Education.

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PREFACE

The Ministry of Education (MoE) through the State Department for Basic Education, has been implementing the Competency Based Education (CBE) since the adoption of the *Basic Education Curriculum Framework* (KICD, 2017) during the national convention in January 2017. The Kenya Institute of Curriculum Development (KICD) developed Competency Based Curriculum (CBC) designs that were used for pilot studies in 2017 and 2018 before the roll-out at Pre-Primary and Primary Early Year (Grades 1 to 3 in 2019). The curriculum was then introduced in the subsequent grades through a *phase in –phase out* strategy, to ensure a smooth transition. Currently, the first cohort of CBC learners are at Grade 9 – the last grade at Junior School. The implementation of CBE has since benefited immensely from the recommendations of the *Presidential Working Party on Education Reforms* (PWPER) that were released in August, 2023 and fully implemented, in terms of curriculum designs.

Grade 10 is the entry class of the Senior School (SS) in Basic Education, a very critical level where learners shall have the opportunity to focus on pathways, tracks and subjects of their choices guided by the potentials demonstrated at earlier levels. The Senior School shall; therefore, form a foundation for further education and training and/or initial competencies for the Industry. This is an important step in the realisation of the Vision and Mission of the curriculum reforms as enshrined in the Sessional Paper No. I of 2019 whose title is: *Towards Realizing Quality, Relevant and Inclusive Education and Training for Sustainable Development* in Kenya.

The Grade 10 curriculum designs, like the designs for earlier levels, are anchored on the Kenyan National Goals of Education. They also present essence statements, general and specific expected learning outcomes as well as strands and sub strands for the learning areas. The designs also outline suggested learning experiences, key inquiry questions, core competencies, Pertinent and Contemporary Issues (PCIs), values and suggested assessment.

The Ministry of Education, through the State Department for Basic Education commits to provide resources and relevant guidelines for a smooth implementation of the CBC. It is expected that the designs will guide teachers to effectively facilitate learners to attain the expected learning outcomes for Grade 10 specifically and Senior School in general.

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ACKNOWLEDGEMENT

The Kenya Institute of Curriculum Development (KICD) Act Number 4 of 2013 (Revised 2019) mandates the Institute to develop curricula, programmes and curriculum support materials for Basic and Tertiary Education and Training. The curriculum development process involves research, international benchmarking and robust stakeholder engagement. Through a systematic and consultative process, the KICD conceptualised the Competency Based Curriculum (CBC) as captured in the *Basic Education Curriculum Framework* (KICD,2017), that responds to the demands of the 21st Century, the United Nations Sustainable Development Goals (SDGs). African Agenda 2063, East African Community Protocol and the aspirations captured in the Kenya Constitution 2010 and the Kenya Vision 2030.

KICD receives its funding from the GoK to enable the successful achievement of the stipulated mandate and implementation of the Government and Education sector plans. The Institute also receives support from development partners targeting specific programmes. The Grade 10 curriculum designs were developed with the support of the World Bank and GoK through the *Kenya Primary Education Equity in Learning Programme* (KPEELP); a project coordinated by the MoE. Therefore, the Institute is very grateful to the GoK, through the MoE and the development partners (especially The World Bank) for support in terms of policy guidance, resource provision and capacity building. Specifically, I wish to express special thanks to the Cabinet Secretary-MoE and the Principal Secretary-State Department for Basic Education.

We also wish to acknowledge the KICD curriculum development officers and other staff, all teachers, educators who took part as panellists; the institutions that participated and representatives of various stakeholders for their roles in the development of the Grade 10 curriculum designs. In relation to this, we acknowledge the support of the Chief Executive Officers of the Teachers Service Commission (TSC), the Kenya National Examinations Council (KNEC) and other agencies in the sector for their support in the process of developing these designs.

Finally, we are very grateful to the KICD Council Chairperson and other members of the Council for consistent guidance and oversight. We assure all teachers, parents and other stakeholders that these curriculum designs will effectively guide the implementation of the CBE at Grade 10 and preparation of learners for transition to further education and training or the world of work.

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NATIONAL GOALS OF EDUCATION

Education in Kenya should:

1. Foster nationalism and patriotism and promote national unity

The people of Kenya belong to different communities, races and religions, but these differences need not divide them. They must be able to live and interact as Kenyans. It is a paramount duty of education to help young people acquire this sense of nationhood by resolving conflicts and promoting positive attitudes of mutual respect, to live together in harmony and foster patriotism, and make a positive contribution to the life of the nation.

2. Promote the social, economic, technological, and industrial needs for national development

Education should prepare the youth of the country to play an effective and productive role in the life of the nation.

a) Social Needs

Education in Kenya must prepare children for changes in attitudes and relationships necessary for the smooth progress of a rapidly developing modern economy. Rapid modernisation is bound to lead to a silent social revolution, and education should assist our youth in adapting to this change.

b) Economic Needs

Education in Kenya should produce citizens with the skills, knowledge, expertise, and personal qualities required to support a growing economy. Kenya is building a modern and independent economy that needs an adequate and relevant domestic workforce.

c) Technological and Industrial Needs

Education in Kenya should provide learners with the necessary skills and attitudes for industrial development. Kenya recognises the rapid industrial and technological changes taking place, especially in the developed world. We can only contribute to this development if our education system is deliberately focused on the knowledge, skills and attitudes that will prepare our young people for these changing global trends.

3. Promote individual development and self-fulfillment

Education should provide opportunities for the fullest development of individual talents and personality. It should help children to develop their potential interests and abilities. A vital aspect of individual development is the building of character.

4. Promote sound moral and religious values.

Education should provide for the development of knowledge, skills and attitudes that will enhance the acquisition of sound moral values and help children to grow up into self-disciplined, self-reliant and integrated citizens.

5. Promote social equality and responsibility.

Education should promote social equality and foster a sense of social responsibility within an education system that provides equal educational opportunities for all. It should give all children varied and challenging opportunities for collective activities and corporate social service irrespective of gender, ability or geographical environment.

6. Promote respect for and development of Kenya's rich and varied cultures.

Education should instill in the youth of Kenya an understanding of past and present cultures and their valid place in contemporary society. Children should be able to blend the best of traditional values with the changing requirements that must follow rapid development to build a stable and modern society.

7. Promote international consciousness and foster positive attitudes towards other nations.

Kenya is part of the international community. It is part of the complicated and interdependent network of peoples and nations. Education should therefore lead the youth of the country to accept membership of this international community with all the obligations and responsibilities, rights and benefits that this membership entails.

8. Promote positive attitudes towards good health and environmental protection.

Education should inculcate in young people the value of good health for them to avoid indulging in activities that will lead to physical or mental ill health. It should foster positive attitudes towards environmental development and conservation. It should lead the youth of Kenya to appreciate the need for a healthy environment.

THE SENIOR SCHOOL IN THE COMPETENCY BASED CURRICULUM

Senior School is the fourth level of Basic Education in the Competency Based Curriculum (CBC) that learners shall proceed to after the Pre-Primary School, Primary School and Junior School levels. The essence of Senior School is to offer learners a pre-tertiary/university/ pre-career experience in which the learners have an opportunity to choose pathways, tracks and subjects where they shall have demonstrated interest and/or potential at the earlier levels. Senior School comprises three years of education for learners generally in the 16-18 years age bracket and lays the foundation for further education and training at the tertiary level and the world of work. In the CBC vision, learners exiting this level are expected to be *engaged, empowered and ethical citizens* ready to participate in the socio-economic development of the nation.

At this level, learners shall take seven (7) learning areas as recommended by the *Presidential Working Party on Educational Reforms* (PWPER, 2023). These shall comprise **four compulsory learning areas** namely: English, Kiswahili, Essential Mathematics/Core Mathematics and Community Service Learning (CSL). Learners will choose between Core Mathematics (for learners pursuing the Science Technology Engineering and Mathematics Pathway) and Essential Mathematics (for learners pursuing the non-STEM pathways). The learner will select three learning areas depending on their career choice, aptitude, interest and personality with guidance by the career teacher at Senior School.

LEARNING OUTCOMES FOR SENIOR SCHOOL

By the end of Senior School, the learner should be able to:

1. Communicate effectively and utilise information and communication technology across varied contexts.
2. Apply mathematical, logical and critical thinking skills for problem solving.
3. Apply basic research and scientific skills to manipulate the environment and solve problems.
4. Exploit individual talents for leisure, self-fulfilment, career growth, further education and training.
5. Uphold national, moral and religious values and apply them in day-to-day life.
6. Apply and promote health care strategies in day-to-day life.
7. Protect, preserve and improve the environment for sustainability.
8. Demonstrate active local and global citizenship for harmonious co-existence.
9. Demonstrate appreciation of diversity in people and cultures.
10. Manage pertinent and contemporary issues responsibly.

LIST OF SUBJECTS AT SENIOR SCHOOL

Core Subjects	Arts & Sports Science	Social Sciences	Science, Technology, Engineering & Mathematics (STEM)
1. English 2. Kiswahili/KSL 3. Core Mathematics/ Essential Mathematics 4. Community Service Learning (CSL)	5. Sports and Recreation 6. Music and Dance 7. Theatre and Film 8. Fine Arts	9. Literature in English 10. Indigenous Languages 11. Fasihi ya Kiswahili 12. Sign Language 13. Arabic 14. French 15. German 16. Mandarin Chinese 17. Christian Religious Education 18. Islamic Religious Education 19. Hindu Religious Education 20. Business Studies 21. History and Citizenship 22. Geography	23. Biology 24. Chemistry 25. Physics 26. General Science 27. Agriculture 28. Computer Studies 29. Home Science 30. Aviation 31. Building Construction 32. Electricity 33. Metalwork 34. Power Mechanics 35. Wood Technology 36. Media Technology 37. Marine and Fisheries Technology

Note:

- Physical Education (PE) and Information Communication and Technology (ICT) will be offered to all learners to facilitate learning, enjoyment and acquisition of life skills.
- Every school shall offer Pastoral/Religious Programme of Instruction (P/RPI) to enhance moral, spiritual and character development of the learners.
- There shall be One lesson for personal or group study to deepen knowledge and enhance skills in areas of choice by the learners.

Table 2 below provides guidelines on distribution of lessons at Senior School.

LESSON DISTRIBUTION AT SENIOR SCHOOL

Subjects	No of Lessons Per Week (40 minutes per lesson)
Core Subjects 4	
1. English	5
2. Kiswahili	5
3. Essential Mathematics/Core Mathematics	5
4. Community Service Learning	3
Elective Subjects	
5. Elective 1	5
6. Elective 2	5
7. Elective 3	5
* <i>Physical Education</i>	3
* <i>ICT Skills</i>	2
* <i>PPI</i>	1
* <i>Learner Personal /Group Study</i>	1
Total Number of Lessons	40

ESSENCE STATEMENT

Agriculture is the Science and art of growing crops and raising animals for food, fiber, medicinal extracts, among other products and services with the aim of sustaining human life. Its inclusion in the school curriculum is anchored on the United Nations Sustainable Development Goal number 2 on zero hunger. Learners taking Agriculture in Senior School will build on the agricultural competencies attained in Junior School. The Agriculture curriculum targets development of competencies in crop production, animal production, agricultural technologies and entrepreneurship applicable in varied social-economic and environmental contexts.

The curriculum will be implemented through learner-centered pedagogies such as experiential learning, collaborative learning, cooperative learning, problem-based learning and project based learning. The curriculum prepares the learner for agricultural related education and training in middle level colleges and university for career development. The acquired skills will also be applied in life-long contexts to address food and nutrition security, employment, environmental conservation and mitigation of climate change.

The curriculum forms foundation for learners to pursue careers in farm management, agri-business management, horticulture, agricultural economics, agricultural engineering, agriculture biotechnology, agriculture education and extension, agronomy, soil science, plant science, and animal science related careers such as animal health, veterinary medicine, animal production and food and dairy technology. Additionally, the learner has an opportunity to develop skills for entrepreneurship in agriculture value chain.

GENERAL LEARNING OUTCOMES

By the end of Senior School, the learner should be able to:

1. Apply agricultural principles in logical and critical thinking skills to solve contextual challenges in the society.
2. Apply basic research and scientific skills to manipulate the environment and solve contemporary agricultural problems.
3. Employ individual talents to utilise agricultural resources for leisure, career and economic growth, further education and training.
4. Use ICT and production technologies effectively in varied agricultural processes.
5. Apply and promote healthy interventions in environmental care through agricultural practices.
6. Protect, preserve and improve the environment for agriculture and socio-economic sustainability.
7. Manage pertinent and contemporary challenges responsibly through agriculture and environmental interventions.

SUMMARY OF STRANDS AND SUB-STRANDS

Strands	Sub-Strands
1.0 Crop Production	1.1 Agricultural Land
	1.2 Properties of Soil
	1.3 Land Preparation
	1.4 Field Management Practices
	1.5 Growing Selected Crops
	1.6 Crop Protection: Weed control
	1.7 General Crop Harvesting
2.0 Animal Production	2.1 Breeds of Livestock
	2.2 Safe Handling of Animals
	2.3 General Animal Health
	2.4 Beekeeping
	2.5 Animal Rearing Project
3.0 Agricultural Technologies and Entrepreneurship	3.1 Tools and Equipment
	3.2 Product Processing and Value Addition
	3.3 Establishing Agricultural Enterprise
	3.4 Marketing Agricultural Produce
	3.5 Composting Techniques

STRAND 1.0: CROP PRODUCTION

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Crop Production	1.1 Agricultural Land	By the end of the sub-strand the learner should be able to: - describe ways of accessing land for agricultural use, - evaluate the utility of land for agricultural production purposes, - analyse natural factors that determine the productivity of land in agriculture, - appreciate the importance of land ownership security in agricultural production.	Learner is guided to: - discuss with a resource person legal ways of accessing and owning land for agricultural use (leasing, inheritance/succession, settlement programmes by the government, allocation by the government, land adjudication, purchase (transfer) and donation) - to take an excursion in the community to study and assess different forms of land and discuss the possible utilities of the land - use digital devices to search for information on natural factors that determine the productivity of land such as climate, altitude, soil factors, topography and biotic factors - make class presentations on the importance of tenure security	- How is land productivity determined for agriculture? - Why is land put to different agricultural uses?

			(security of land ownership) in agricultural production.	
Core Competencies to be developed: • Critical thinking and Problem-solving: learner studies and assesses different forms of land and their possible utilities for agricultural purposes. • Communication and Collaboration: learner speaks clearly and effectively while discussing ways of accessing land and the importance of security of ownership in agricultural production.				
Values: • Respect: learner displays humility while discussing ways of accessing land for agricultural production. • Integrity: learner develops honesty while discussing with a resource person legal ways of accessing land for crop production.				
Pertinent and Contemporary Issues (PCIs): • Socio-economic and Environmental issues: learner appreciates the importance of environmental awareness when analysing natural factors that determine the productivity of agricultural land. • Consumer Awareness: learner learns legal ways of land ownership and associated land rights to prevent fraud.				

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Crop Production	1.2 Properties of soil	By the end of the sub-strand the learner should be able to: - investigate components of soil that influence crop production, - examine properties of a soil for crop production, - describe properties of soil that influence crop production, - investigate selected properties of soil that influence crop production, - relate the importance of soil profile to crop production, - acknowledge the importance of soil properties in crop production.	Learner is guided to: - discuss components of soil (mineral particles, organic matter, water, and air) and their importance in crop production - carry out simple experiments using soil samples to examine the various components of soil (mineral particles, organic matter, water and air) - discuss how physical properties (<i>soil texture, soil structure, porosity, permeability</i>), chemical properties (<i>soil pH</i>), and biological properties (<i>living organisms in the soil</i>) influence crop production - carry out experiments to test properties of soil (<i>soil texture, water holding capacity, soil capillarity, and soil pH</i>). - take field excursion along roadsides and excavations to observe soil	- How do soil properties influence crop production? - How can we investigate various properties of soil?

			profile and discuss its importance in crop production. • use digital and non-digital resources to search for the importance of physical, chemical and biological properties of soil in crop production.	
Core Competencies to be developed: • Critical thinking and Problem-solving: researching skills as learner conducts experiments to test various soil properties. • Learning to Learn: learner seeks information on how soil profile affects crop production during field excursions.				
Values: • Unity: learner cooperates with others while discussing and making presentations on factors that influence soil properties during soil formation process. • Peace: learner resolves differences amicably with others when conducting experiments on investigation of soil properties.				
Pertinent and Contemporary Issues (PCIs): • Social economic and environmental issues: learner appreciates the importance of soil properties, for example, in crop production, construction of buildings, and construction of safe road infrastructure. • Online safety: learner demonstrates responsible use of the internet when searching for information on the importance of various soil properties in crop production.				

Strand	Sub-strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Crop Production	1.3 Land Preparation • <i>Land clearing</i> • <i>Primary cultivation</i> • <i>Secondary cultivation</i> • <i>Tertiary operations.</i> • <i>Conservation tillage</i>	By the end of the sub-strand the learner should be able to: a) describe activities of fallow land preparation to appropriate seedbed, b) justify use of conservation tillage practices in crop production c) carry out land preparation operations for a selected crop, d) appreciate the importance of land preparation in crop production.	Learner is guided to: • discuss the activities carried out on fallow land (land clearing, primary cultivation, secondary cultivation, and tertiary operations) to prepare appropriate seedbeds • discuss how conservation tillage practices (zero tillage, minimum tillage) are carried out and justify their adoption in crop production • carry out fallow land in preparation activities applicable for the establishment of a selected crop • make presentations on the importance of land preparations in crop production.	How does proper land preparation contribute to crop production?
Core Competencies to be developed: • Communication and Collaboration: learner recognises the ideas of others when discussing land preparation activities and presenting the importance of land preparation. • Self-efficacy: learner examines fallow land and successfully carries out applicable land preparation activities for a selected crop.				

Values:

- Peace: learner resolves conflicts when sharing tools and tasks with others during land preparation activities.
- Responsibility: learner offers leadership and guidance to others harmonious working as learners carry out land preparation activities.

Pertinent and Contemporary Issues (PCIs):

- Social economic and environmental issues: learner observes safety of self and others when working with tools during land preparation activities.
- Social economic and environmental issues: learner explains how conservation tillage helps to conserve the environment in crop production.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Crop Production	1.4 Field Management Practices • <i>Vegetable pruning</i> • <i>Pruning perennial crop</i> • <i>Top dressing</i>	By the end of the sub-strand the learner should be able to: a) describe management practices of selected vegetable and perennial crops, b) carry out selected management practices in crop production, c) explain factors considered in top-dressing a crop field, d) appreciate the importance of selected management practices in crop production.	Learner is guided to: • search for information or take a field trip to study the pruning of vegetables such as capsicum, tomatoes, and perennial crops such as bananas, pyrethrum, coffee, and tea • carry out pruning of selected vegetable crops such as tomatoes and capsicum • explore methods of top-dressing fertilisers (broadcasting, side dressing and foliar application) • discuss factors considered (proper timing, type and form of fertiliser, top-dressing method, correct amount of fertiliser, stage of crop growth) while top-dressing selected crop field • discuss and present on the importance of pruning and top dressing in crop production.	How do field management practices influence crop production?

Core Competencies to be developed:

- Communication and Collaboration: learner demonstrates effective speaking and listening skills when discussing and presenting on the importance of carrying out field management practises.
- Critical thinking and Problem-solving: learner follows instructions when carrying out pruning and top dressing crops.

Values:

- Love: learner portrays a caring attitude towards other learners when carrying out field management practices.
- Responsibility: learner demonstrates self-drive when carrying pruning of selected vegetable crops.

Pertinent and Contemporary Issues (PCIs):

- Socio-economic and Environmental issues: learner observes safety precautions while handling top dressing fertilisers.
- Socio-economic and Environmental issues: learner appreciates the importance of proper timing when applying top dressing fertilisers to ensure absorption.

Strand	Sub-strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Crop Production	1.5 Growing Selected Crop <i>Nursery bed establishment</i> <i>Nursery bed management</i> <i>Crop establishment</i> <i>Field management practices.</i>	By the end of the sub-strand the learner should be able to: - determine crops that are established through a nursery, - grow a selected crop by raising it from a nursery bed, - justify management practices for a selected crop.	Learner is guided to: - brainstorm to determine appropriate crops such as vegetables or any other crop that are established from the nursery - establish a crop from a nursery bed, transplant to the seedbed and carry out appropriate management practices during its growth cycle - discuss and make class presentations on why field management practices are carried out on a selected crop.	- How do we establish a crop from a nursery? - How do management practices influence crop productivity?
Core Competencies to be developed: - Learning to Learn: learner orders and prioritises tasks when carrying out field management practices on crops to ensure successful crop production. - Self-efficacy: learner develops self-confidence and focus when growing and managing a crop to achieve the intended goals.				
Values - Unity: learner collaborates with others when establishing and managing crops. - Social justice: learner fosters inclusivity and non-discrimination when discussing and making class presentations.				

Pertinent and Contemporary Issues (PCIs):

- Social economic and environmental issues: learner upholds gender equity and inclusion by according other learners the opportunity to participate in the crop growing project.
- Social economic and environmental issues: learner observes safety and security of self and others when carrying out activities on the establishment and management of selected crops.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Crop Production	1.6 Crop Protection: Weed control	By the end of the sub-strand the learner should be able to: a) identify weeds in a crop field, b) classify weeds based on the provided criteria, c) examine various methods of weed control, d) carry out weed control using appropriate methods, e) appreciate the economic importance of weeds to a farming household.	Learner is guided to: • take an excursion to identify weeds in a crop field and make herbarium on identified weeds • use digital and non-digital resources to classify weeds based on morphology and growth cycle • discuss various methods of weed control (<i>physical/mechanical; cultural; biological; chemical; and legislative methods</i>) • carry out weed control in a crop field using appropriate methods (<i>physical; cultural; biological; chemical; and integrated methods</i>) • discuss and make class presentations on the pros and cons of weeds to a farming household.	1. How do weeds affect crop production? 2. Why is weed control done in crop production?
Core Competencies to be developed: • Digital Literacy: learner uses digital technology when searching for information on how weeds are classified. • Creativity and imagination: learner uses innovative ideas when making a herbarium for weed identification purposes.				

Values:

- Responsibility: learner develops a sense of obligation when carrying out weed control in a crop field.
- Respect: learner demonstrates regard for others when discussing making presentations on pros and cons of weeds to a farming household.

Pertinent and Contemporary Issues (PCIs):

- Health promotion issues: learner discusses the effects of irresponsible chemical weed control on human health and how this can be minimised.
- Socio-economic and Environmental issues: learner is sensitised on the positive and negative effects of weeds in an agricultural environment.
- Learner support programme: learner seeks guidance services and support when making herbarium and controlling weeds.

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
1.0 Crop Production	1.7 General Crop Harvesting <i>Factors determining harvesting</i> <i>Pre-harvesting practices</i> <i>Harvesting methods</i> <i>Post-harvest practices</i>	By the end of the sub-strand the learner should be able to: - explain factors that determine the harvesting of a crop produce, - carry out the harvesting processes for selected crop produce, - acknowledge the importance of harvesting processes in crop production.	Learner is guided to: - search for information using digital devices on factors such as harvest timing, stage of growth and purpose, that determine the harvesting of crop produce - discuss the harvesting process (<i>pre-harvest practices, harvesting and post-harvest practices</i>) for <i>tuber and cereal crops</i> - carry out harvesting processes (<i>pre-harvest practices, harvesting and post-harvest practices</i>) for <i>tuber and cereal crops</i> - discuss with a resource person the importance of pre-harvest, harvest and post-harvest processes in production tuber and cereal crops.	How does the harvesting process affect crop production?
Core Competencies to be developed: - Digital Literacy: learner uses the internet when searching for information on factors that determine harvesting of crop produce. - Critical thinking and Problem-solving: learner uses skills of observation, evaluation, and decision-making to determine the appropriate harvesting processes for tuber and cereal crops.				

Values:

- Integrity: learner exhibits honesty when determining the appropriate time to harvest high-quality tuber and cereal crops.
- Unity: learner takes part in conversations and activities on the harvesting process of tuber and cereal crops.

Pertinent and Contemporary Issues (PCIs):

- Socio-economic and Environmental issues: learner discusses how to ensure food safety, for example, by properly storing cereals to prevent aflatoxin poisoning, which is dangerous to human health.
- Socio-economic and Environmental issues: learner observes safety precautions and practices when using digital devices to search for factors that determine the harvesting of crop produce.

Suggested Assessment Rubric

Level Indicator	Exceeds Expectations	Meets Expectations	Approaches Expectations	Below Expectations
Ability to describe ways of accessing land for agricultural use	The learner describes more than four ways of accessing land for agricultural production.	The learner describes four ways of accessing land for agricultural production.	The learner describes two to three ways of accessing land for agricultural production.	The learner describes less than two ways of accessing land for agricultural production.
Ability to analyse natural factors that determine the productivity of land in agriculture	The learner analyses more than four natural factors that determine the productivity of land in agricultural production.	The learner analyses four natural factors that determine the productivity of land in agricultural production.	The learner analyses two to three natural factors that determine the productivity of land in agricultural production.	The learner analyses less than two natural factors that determine the productivity of land in agricultural production.
Ability to investigate various properties of	The learner creatively investigates four	The learner investigates four	The learner investigates two to	The learner investigates less than

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Agricultural Technologies and Entrepreneurship	3.3 Establishing Agricultural Enterprise	By the end of the sub-strand the learner should be able to: - explain the factors of production in an agricultural enterprise, - describe ways of acquiring capital to establish an agricultural enterprise, - examine factors to consider in selecting an agricultural enterprise, - appreciate the role of various factors of production in establishing an agricultural enterprise.	Learner is guided to: - discuss the factors of production (<i>land/space, labour, entrepreneurship, capital</i>) in an agricultural enterprise - discuss with a resource person alternative ways such as borrowing, savings, disposing-off assets, grants, and donations of acquiring capital to establish an agricultural enterprise - use digital and non-digital resources to search and determine factors to consider such as physical infrastructure, inputs, labour requirements, skill requirement, production techniques, legal requirements and market in selecting an agricultural enterprise - discuss and present the role of various factors of production in	How do we establish an agricultural enterprise?

			establishing an agricultural enterprise.	
Core Competencies to be developed: • Critical thinking and Problem-solving: learner develops research skills when searching and determining factors to consider in selecting an agricultural enterprise and when evaluating appropriate sources of support services for an agricultural enterprise • Creativity and Imagination: learner connects and relates ideas about mobilising capital when establishing an agricultural enterprise.				
Values: • Integrity: learner develops honesty when discussing legal or acceptable ways of mobilising capital to establish an agricultural enterprise. • Respect: learner understands and appreciates others when discussing and presenting factors of production in an agricultural enterprise.				
Pertinent and Contemporary Issues (PCIs): • Financial literacy: learner develops investment skills when discussing with a resource person ways of mobilising capital for establishing agricultural enterprise. • Consumer education: learner develops consumer awareness when evaluating various ways of mobilising capital to establish an agricultural enterprise, for example, the importance of reading the terms and conditions of loan agreements from financial institutions carefully to understand their obligations.				

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Agricultural Technologies and Entrepreneurship	3.4 Marketing agricultural produce	By the end of the sub-strand the learner should be able to: - describe ways of preparing agricultural produce for marketing, - prepare agricultural produce for marketing, - discuss market outlets for agricultural produce, - evaluate expenses incurred in marketing agricultural produce, - appreciate the importance of preparing agricultural produce for marketing.	Learner is guided to: - discuss ways of preparing agricultural produce for marketing such as weighing, sorting, grading, packaging, branding, and labeling. - visit an agricultural market outlet to observe and learn how different agricultural produce are weighed, sorted, graded, packaged, branded, labelled, and displayed; and prepare any applicable produce in school. - discuss various market outlets for agricultural produce (<i>digital platforms and physical market outlets</i>) - inquire from a resource person about expenses incurred in marketing activities such as <i>transportation costs, advertisement costs, market authority charges, and taxes</i>	How can we prepare agricultural produce for the market?

			share experiences on the importance of preparing agricultural produce for marketing (<i>weighing, sorting, grading, packaging, branding, and labeling</i>).	
Core Competencies to be developed - Self-efficacy: learner develops self-awareness and planning skills when demonstrating the preparation of vegetable crop produce for marketing and displaying to the class. - Communication and Collaboration: learner develops effective writing skills when demonstrating how to prepare (through branding and labelling) and display agricultural produce for marketing.				
Values - Integrity: learner develops honesty by demonstrating proper sorting and grading, weighing, labelling, branding, and fair pricing when marketing agricultural produce and observing ethics when using online marketing platforms. - Love: learner respects a resource person when inquiring about expenses incurred in marketing activities.				
Pertinent and contemporary issues - Financial literacy: learner learns how to earn income after learning how vegetable crop produce is prepared, displayed and sold in the market, and how to determine if marketing of agricultural produce is financially worthwhile after taking into account all the expenses incurred. - Social-economic and environmental issues: learner develops netiquette skills when discussing online marketing platforms for agricultural produce. - Social-economic and environmental issues: learner develops skills in safe road use by walking on the walkways and using designated crossing points when visiting a market to observe how agricultural produce is prepared for marketing.				

Strand	Sub-Strand	Specific Learning Outcomes	Suggested Learning Experiences	Suggested Key Inquiry Question(s)
3.0 Agricultural Technologies and Entrepreneurship	3.5 Composting Techniques	By the end of the sub-strand the learner should be able to: - describe composting methods in the production of organic manure, - examine factors that influence the quality of compost manure, - carry out conventional composting methods for the production of organic manure, - carry out innovative composting methods for the production of organic manure, - appreciate the role of composting in soil improvement.	Learner is guided to: - use ideas from digital and non-digital resources to describe methods of composting (<i>conventional methods; innovative methods</i>) using locally available resources at the farm level. - discuss and present factors that influence the quality of compost manure such as materials used, the process of composting, and storage conditions - carry out composting with guidance of a resource persons or other sources to produce organic manure through the four-pit method - use digital resources to acquire ideas and carry out innovative composting through techniques such as <i>vermicomposting and containerised composting</i> - utilise the compost manure on existing crop enterprises to	Why is composting relevant to soil improvement?

			appreciate the role of composting in soil improvement.	
Core Competencies to be developed: • Learning to learn: learner finds out how to carry out innovative composting techniques and establish the qualities of well decomposed compost manure. • Self-efficacy: learner keeps trying to follow the learnt composting procedure until quality compost is produced.				
Values: • Responsibility: learner exhibit leadership skills in carrying out the composting process using conventional and innovative methods. • Love: learner demonstrates self-sacrifice by looking for materials for composting and when managing the composting process to produce quality manure.				
Pertinent and Contemporary Issues (PCIs): • Financial literacy: learner develops skills of saving money by learning how to make compost manure for soil improvement using locally available resources instead of buying chemical fertilisers. • Socio-economic and Environmental issues: learner conserves soil by applying compost manure to an existing crop enterprise.				

Suggested Assessment Rubric

Level Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectations	Below Expectations
Ability to carry out various agricultural tasks using appropriate tools and equipment.	Learner is able to carry out all agricultural tasks using appropriate tools and equipment.	Learner is able to carry out most agricultural tasks using appropriate tools and equipment.	Learner is able to carry out a few agricultural tasks using appropriate tools and equipment.	Learner is able to carry out a few agricultural tasks using appropriate tools and equipment with a lot of support.
Suggest methods of value addition for selected agricultural produce.	Learner suggests methods of value addition for more than seven selected agricultural produce.	Learner suggests methods of value addition for seven selected agricultural produce.	Learner suggests methods of value addition for three to six selected agricultural produce.	Learner suggests methods of value addition for less than three selected agricultural produce.
Ability to analyse ways of acquiring capital to establish an agricultural enterprise.	Learner critically analyses five ways of acquiring capital to establish an agricultural enterprise.	Learner analyses four ways of acquiring capital to establish an agricultural enterprise.	Learner analyses two to three ways of acquiring capital to establish an agricultural enterprise.	Learner analyses less than two ways of acquiring capital to establish an agricultural enterprise.
Ability to discuss market outlets for agricultural produce. (digital and physical market outlets).	Learner discusses, in depth, market outlets for agricultural produce giving varied examples.	Learner discusses market outlets for agricultural produce.	Learner partially discusses market outlets for agricultural produce.	Learner partially discusses market outlets for agricultural produce only when given a lot of support.

Level Indicator	Exceeds Expectation	Meets Expectation	Approaches Expectations	Below Expectations
Ability to describe methods of composting in the production of organic manure.	Learner describes, in detail, methods of composting in the production of organic manure.	Learner describes methods of composting in the production of organic manure.	Learner describes methods of composting in the production of organic manure but leaves out a few details.	Learner describes methods of composting in the production of organic manure but leaves out many details.

APPENDIX 1: LIST OF ASSESSMENT METHODS, LEARNING RESOURCES AND NON-FORMAL ACTIVITIES

Strand	Sub-Strand	Suggested assessment methods	Suggested learning resources	Suggested non-formal activities
1.0 Crop Production	1.1 Agricultural Land	• Written test • Written assignment • Observation of learning activities.	Relevant video clips and pictures on factors that determine land productivity. Manilla papers and marker pens to make presentations on the importance of land in agricultural production.	Learners to develop posters to sensitise members of the school community on legal procedures of land acquisition.
	1.2 Properties of soil	• Written assignments • Observation of learning activities	Photographs of different soil textural classes. Varied soil samples. Soil testing materials and equipment such as soil testing kits and PH indicators. Soil sampling tools such as <i>jembe</i>, auger, <i>panga</i>, and soil containers.	Learners to develop an item in soil testing during a science and technology forums. Learners observe the soil profile on the road sides.

Strand	Sub-Strand	Suggested assessment methods	Suggested learning resources	Suggested non-formal activities
	1.3 Land Preparation	• Observation of learning activities • Written test • Oral assessment	School farm Farm tools and equipment such as <i>jembe</i> , fork <i>jembe</i> , panga, and slasher	Develop a demonstration plot on well-prepared land at school or home set up. Initiate and participate in Young Farmers Club on land preparation activities
	1.4 Field Management Practices • Vegetable pruning • Pruning perennial crop • Top-dressing	• Observation of pruning and top dressing activities • Written test •	Pruning tools such as secateurs, pruning saw, knife, and sheers among others. Top-dressing tools and materials such as knapsack/ hand sprayer, gloves, and samples of appropriate fertilizers.	Learner participates in young farmers' activities in pruning and top-dressing activities. Learners participate in making posters on the importance of the application of inorganic fertilizers
	1.5 Growing Selected Crop • Nursery bed establishment • Nursery bed management	• Written test • Observation of nursery management practices or	School farm, innovative gardens, applicable tools and equipment such as rake, garden trowel, watering can	Learner to participate in club activities in nursery bed establishment, management.

Strand	Sub-Strand	Suggested assessment methods	Suggested learning resources	Suggested non-formal activities
	• Crop establishment • Field management • Harvesting	- harvesting activities • Project reports		Learner to participate in nursery management activities at home.
	1.6 Crop Protection: Weed control	• Observation of weed control activities • Project portfolio on weed identification • Oral assessment	Photographs of weeds, weeds herbarium, local environment, video clip on weed and weed control methods, samples of appropriate herbicides and tools, knapsack/ hand sprayer.	Learner to participate in activities to sensitise the school community on the economic importance of weeds.
	1.7 General Crop Harvesting • Factors determining harvesting • Post harvesting practices • Harvesting methods • Post-harvest practices	• Observation of activities such as harvesting • Written tests • Project reports	Crop in a school farm, innovative gardens, photographs of crops for harvesting.	Engage in harvesting a cereal or tuber crops.

Strand	Sub-Strand	Suggested assessment methods	Suggested learning resources	Suggested non-formal activities
2.0 Animal Production	2.1 Breeds of Livestock • Cattle • Pigs • Rabbits • Sheep • Goats	• Written test • Oral assessment as learners observe to distinguish different livestock breeds	Live animal breeds, photographs/videos / pictures of different livestock breeds.	Learner to visit an Agricultural Show in the livestock breed section to learn about them.
	2.2 Safe handling of Animals	• Observation of animal handling activities • Written assignment	Animal handling equipment such as bull ring, rope, holster Video clips and pictures on animal handling.	Learners to conduct a sensitisation on handling different animals using the correct equipment.
	2.3 General Animal Health	• Written reports • Oral presentation of signs of ill health	Video clips, photographs of ill animals. Animals in the immediate environment and in school. Relevant resources such as antiseptics, drugs, bolus gun, hypodermic syringes, and needles	Learners to participate in livestock disease control practices in the community such as cleaning livestock structures.

Strand	Sub-Strand	Suggested assessment methods	Suggested learning resources	Suggested non-formal activities
	2.4 Beekeeping	• Written tests • Observation • Oral assessment on safety in bee handling	Hives or hive models, bee handling equipment such as hive tool smoker, bee suit	Learners to participate in public fora to sensitise the community on proper siting of apiary
	2.5 Animal Rearing Project	• Project portfolio • Observation as learners carry out management practices in animal-rearing	Appropriate rearing structures such as rabbit hatch, poultry house, feeders and waterers, and feeding materials	Learners to participate in animal-rearing in their community set-ups
3.0 Agricultural Technologies and Entrepreneurship	3.1 Tools and Equipment	• Observation of learners as they identify and use appropriate tools for agricultural purposes • Oral assessment	Sample agricultural tools such as panga, jembe, pruning knife and burdizzo, elastrator and ring dehorning wire, claw hammer, and hand saw.	Learners to engage in agricultural activities involving the use of tools in the community or at school.
	3.2 Product Processing and Value Addition	• Observe learners as they carry out value-addition and processing activities	Source of heat, cutting tools, crop produce to be processed. (fruits, tubers and nuts).	Participate in the proper storage, value addition and preservation of agricultural produce in school and at home

Strand	Sub-Strand	Suggested assessment methods	Suggested learning resources	Suggested non-formal activities
		• Project report		
	3.3 Establishing Agricultural Enterprise	• Written test • Oral assessment • Project report as learners establish an agricultural enterprise	Established enterprises in school or its environs, resource person,	Initiate micro-enterprises at home or in school clubs.
	3.4 Marketing agricultural produce	• Written test • Observe practical activities as they prepare and market agricultural produce • Oral assessment	Sample agricultural produce such as vegetable, cereals, pulses milk, eggs, chicks, the local market	Learners to visit market outlets to observe how agricultural produce is prepared for marketing.
	3.5 Composting Techniques	• Written test • Oral assessment • Observation of the composting process	Local composting materials of plant and animal origin, pictures, video clips and illustrations, and samples of compost manure.	Learners to advocate for the use of compost manure at home and in school through messages on posters.

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APPENDIX II: WEEDS TO BE STUDIED

COMMON NAME

1. Black Jack
2. Mexican marigold
3. Oxalis/sorrel
4. Double thorn
5. Thorn apple
6. Couch grass
7. Nut grass
8. Wandering Jew
9. Sow thistle
10. Devil's horsewhip
11. Macdonaldeye/gallant soldier
12. Sodom apple
13. Black nightshade
14. Chinese lantern
15. Bracken fern
16. Love grass/Bristly foxtail
17. Cleavers
18. Stinging nettle
19. Fat hen/Goosefoot
20. Rape Weed
21. Wild Oats

BOTANICAL NAME

Bidens pilosa
Tagetes minuta
Oxalis species
Oxygonum sinuatum
Datura stramonium
Digitaria scalarum
Cyperus rotundas
Commelina bengalensis
Sonchus oleraceus
Achyranthes aspera
Gallinsoga parviflora
Solanum incanum
Solanum nigrum
Nicandra physalodes
Pteridium aquillinum
Seteria verticillata
Gallium spurium
Utica massaica
Chenopodium species
Brassica napus
Avena fatua

COMMON NAME

- 22. Lantana/Tick berry
- 23. Water hyacinth
- 24. Witchweed
- 25. Creeping indigo

BOTANICAL NAME

- Lantana camara*
- Eichhornia crassipes*
- Striga hermonthica*
- Indigofera spicata*

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